

Calculators OK. Give all probability answers as percentages rounded to 2 decimal places. Round all Expected Value problems to 2 decimal places where appropriate. Show the work that leads to your answer in all problems.

1. The probability of the Golden Gate Bridge being foggy on any day in November is 20%.
 - a) Find the probability that it will **be foggy** at the Golden Gate Bridge all week (7 consecutive days). [2]
 - b) Find the probability that there will be **3 or fewer** foggy days at Golden Gate Bridge this week. [3]
 - c) Suppose there is a week in November where there are **3 or fewer** foggy days at Golden Gate Bridge. What is the probability that there were **exactly 2** foggy days at the Bridge that week? [2]
 - d) What is the expected number of foggy days at the bridge this November (30 days)? [1]
2. Consider a binomial event such that the probability of 12 consecutive failures is 3%. Determine the constant probability of a single success. [3]

3. You can choose to play one of two games. Each game costs one dollar to play.

Game 1:

A wheel with three numbers on it--zero, one, and two--is spun so that there is a 40% chance that the wheel lands on zero, a 20% chance the wheel lands on one, and a 40% chance the wheel lands on two. You win the amount in dollars of the number that the wheel lands on.

Game 2:

A different wheel with three numbers on it--zero, one, and two--is spun so that there is a 5% chance that the wheel lands on zero, a 85% chance the wheel lands on one, and a 10% chance the wheel lands on two. You win the amount in dollars of the number that the wheel lands on.

Analyze both Game 1 and Game 2 mathematically with respect to their Expected Value and **write a sentence about which game you'd prefer to play.** [5]

4. Keno is a game of chance that costs \$2 to play. Here's how it works: you select 4 numbers out of 80. After your selection, 20 of the 80 numbers are randomly chosen as winners. (Note: order of numbers does not matter in Keno). If all 4 of your numbers are winners you get \$100. If 3 of your 4 numbers are winners you get \$35. If 2, 1 or 0 of your numbers are winners you get nothing. What is the expected value of playing 1 game of Keno? [4]